

LLM Benchmark Review — Q1 2026

AI Research Tracker / Internal Reference

Overview

This document reviews benchmark methodology issues identified in recent frontier model evaluations, focusing on MMLU result comparability across evaluation protocols.

Gemini Ultra MMLU — Methodology Clarification

Google's December 2023 technical report claimed Gemini Ultra achieves 90.0% on MMLU, the first model to exceed human expert performance (89.8%). This figure was widely reported as a breakthrough result.

However, the 90.0% score was produced using CoT@32 evaluation: 32 chain-of-thought samples per question with majority voting. This is a fundamentally different protocol from the standard 5-shot direct-answer evaluation used for all other entries in comparative tables, including GPT-4 (86.4%) and Claude 3 Opus (86.8%).

Under the standard 5-shot protocol, Gemini Ultra scores 83.7% — below GPT-4.

This means the 90.0% claim in the existing wiki entry is misleading as a direct comparison to other models. The table entry and the claim that Gemini Ultra was the first model to surpass human expert performance should be qualified with the methodology difference.

Corrected Comparison Table

Model	MMLU Score	Protocol	Human Expert?
GPT-4	86.4%	5-shot	No
Claude 3 Opus	86.8%	5-shot	No
Gemini Ultra	90.0%	CoT@32	Yes (CoT@32 only)
Gemini Ultra	83.7%	5-shot (equiv.)	No
Llama 3 70B	82.0%	5-shot	No

Recommendation

The llm-benchmarks wiki page should be updated to reflect that the Gemini Ultra 90.0% MMLU figure is not directly comparable to 5-shot results from other models. The claim that Gemini Ultra is the first model to surpass human expert performance on MMLU should be qualified accordingly.